1.6 Solving Linear Systems Using Technology

Name _____ Period ____ Date _____

Algebra II 1.6 (30-minute plan) · Student sheet

I can…

- write an augmented matrix for a system and state its dimensions.
- use technology to solve a system of linear equations.
- use technology to solve real-life problems.

Vocabulary · fill in the missing word

Matrix — a _____ arrangement of numbers in rows and columns

Dimensions — m × n: m rows by n _____

Augmented matrix — the _____ and constants of a system, with each equation as a row

Reduced row-echelon form (rref) — the form _____ gives, with 1s on the diagonal and 0s elsewhere; the last column is the solution

Part 1 · Notes and examples (fill in each note, then work the example)

Dimensions: Dimensions are _____ × _____.

Augmented matrix: Write each equation in _____ form first, and use _____ for a missing variable.

Example 1

Write an augmented matrix for the system. Then state its dimensions.

$2x - y + 3z = 1$

$x + 4z = -1$

$-3x + 2y - z = -4$

Reading rref: In rref, the solution is in the _____ column.

Example 2

Use technology to solve the system in Example 1.

Part 2 · Partner practice

Partner practice

Write an augmented matrix and give its dimensions.

$x + y + z = 4$

$2x - z = 3$

$y + 3z = 8$